Vivekanand Education Society's
Institute of Technology

(Autonomous Institute Affiliated to University of Mumbai, Approved by AICTE & Recognised by Govt. of Maharashtra)
NAAC accredited with 'A' grade

Project Logbook

Biotoken: A Decentralized Framework for Reagent provenance and Verification using Zero-knowledge proof and Blockchain

Group Members

Roll No: 45 Name: Khushi Singh

Roll No: 37 Name: Harsh Patil

Roll No: 29 Name: Ishaan Khan

Mentor:

Mrs. Kajal Joseph

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INFORMATION TECHNOLOGY DEPARTMENT

VISION:

To foster innovation, ethics, and lifelong learning in the field of Information Technology, shaping global professionals and leaders for sustainable development.

MISSION:

1. Deliver holistic, learner-centric IT education with societal relevance.
2. Drive research, innovation, and entrepreneurship through collaboration.
3. Uphold ethical, sustainable, and inclusive technology practic
4. Shape globally competent professionals with skills and lifelong learning.

Program Outcome:

PO1	Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop the solution of complex engineering problems.
PO2	Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
PO3	Design/Development of Solutions: Design creative solutions for complex engineering problems and design / develop systems /components / processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)
PO4	Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).
PO5	Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)
PO6	The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).
PO7	Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)
PO8	Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
PO9	Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences
PO10	Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
PO11	Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change.
PSO 1	**Computing & Emerging Technology:** Apply core computing and modern technologies, such as AI, Cloud Computing, IoT, cybersecurity, and data-driven tools, to build efficient and innovative solutions.
PSO 2	**Professionalism, Research & Innovation :** Exhibit professional ethics, teamwork, and research capabilities to create innovative and sustainable technological solutions.

Course Outcomes

Course Outcomes: After successful completion of the course students will be able to:	
1	Identify and define a real-world engineering problem and apply knowledge of computing and engineering fundamentals to propose an appropriate solution.
2	Conduct a systematic literature review and analyze existing solutions to justify the problem statement, objectives, and scope of the project.
3	Elicit and analyze functional and non-functional requirements and prepare a formal problem specification and project plan.
4	Design a high-level system architecture and select appropriate tools, technologies, and methodologies considering technical, societal, and environmental constraints.
5	Work effectively in a team to prepare and present a professional project proposal, design documents, and reviews using appropriate technical communication.
6	Demonstrate awareness of ethical, legal, and professional responsibilities relevant to the proposed project.

CO-PO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	3	2	2	-	2	-	-	-	-	-	-	3	
CO2	2	3	-	3	-	-	-	-	2	-	2	2	2
CO3	2	3	3	-	2	2	-	2	-	3	-	2	-
CO4	2	2	2	2	3	2	-	-	-	2	-	3	-
CO5	-	-	2	-	-	-	-	3	3	2	2	-	3
CO6	-	-	-	-	-	3	3	-	-	-	2	-	3

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Department of Information Technology

2025-2026

Group No: 39

Project Title: Biotoken: A Decentralized Framework for Reagent provenance and Verification using Zero-knowledge proof and Blockchain

Mentor: Prof. Kajal Joseph

Students Details:

	Roll No	Name	Class	Contact	Email	Sign
Member 1	45	Khushi Singh	D15C	9321622644	2023.khushi.singh@ves.	Khushi
Member 2	37	Harsh Patil	D15C	8451890209	2023.harsh.patil@ves	
Member 3	29	Ishaan Khan	D15C	7045546333	2023.mohammad.khan@ves.ac.in	

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Project Progress Report
SEM –VI

Project Group No. 39

Title: Biotoken: A Decentralized Framework for Reagent provenance and Verification using Zero-knowledge proof and Blockchain

Mentor: Prof. Kajal Joseph

Roll No	Name of Project Member
45	Khushi Singh
37	Harsh Patil
29	Ishaan Khan

Week/Date	Work Done	Mentor's Remark	Sign of Mentor
1.	Create the working dev environment. Created Polygon Amoy testnet wallet	Topic finalized and work started	
2.	Deployed the contract address (ERC-721) + ABI	Smart contracts are designed.	
3.	circom ZK circuit for HPLC fingerprint commitment, pk, vk file	ZKP part completed	
4.	Cleaned dataset to build AI model, EDA notebook, normalise spectra, define tolerance thresh	Dataset found and cleaned Working on AI	
5.	Proof π was generated and verified on chain. Connected AI model to ZK proof	POC of AI done	
6.	Build React Frontend, created dashboards for Lab and Manufac -turer (Working Screens).	UI wireframes designed & integrated	